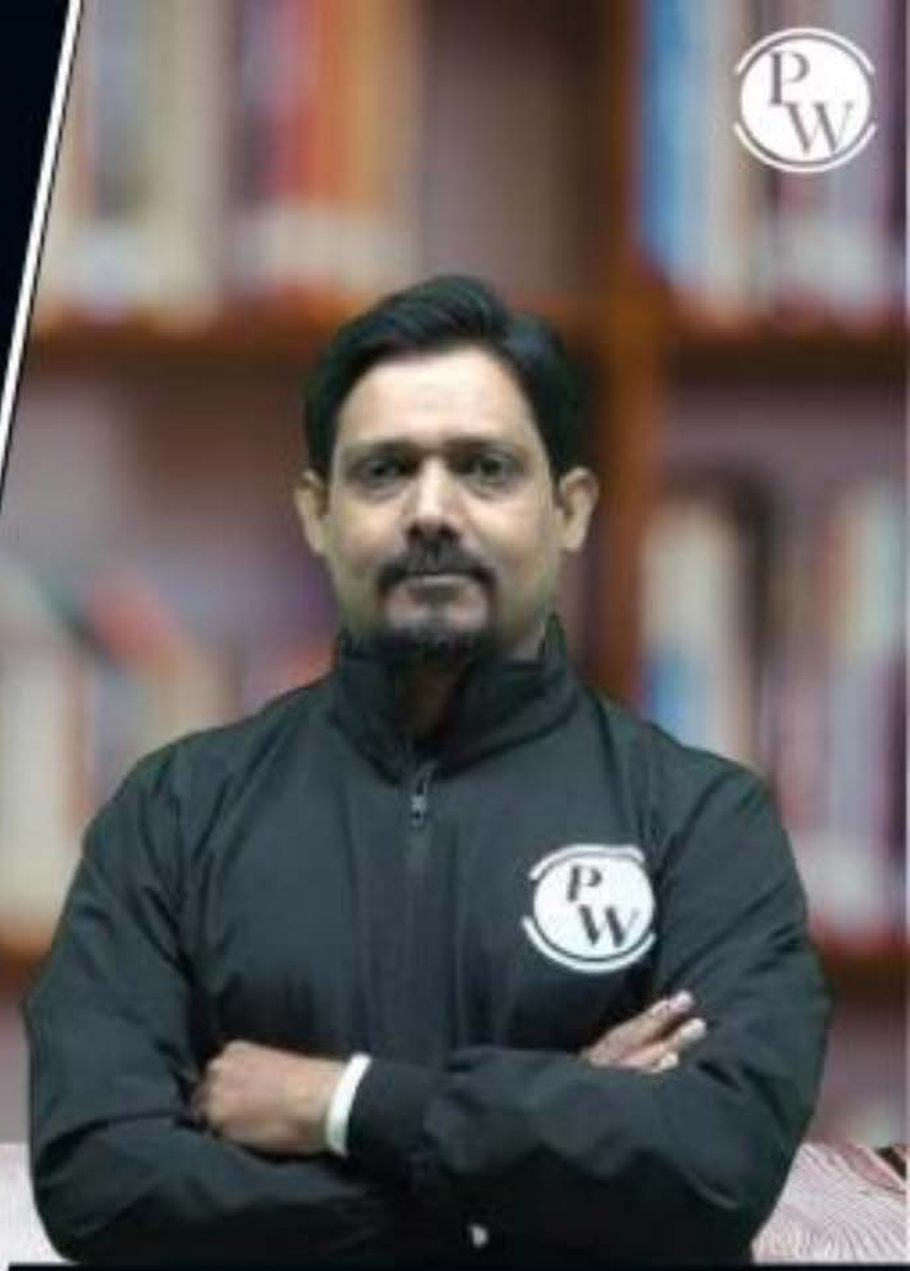
Computer Science & IT



Data Structure & Programming

Tree

Lecture No. 09



By- Abhishek Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Heap Delete

Heapify

Adjust ←

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Algorithm Adjust

Counting

Graph

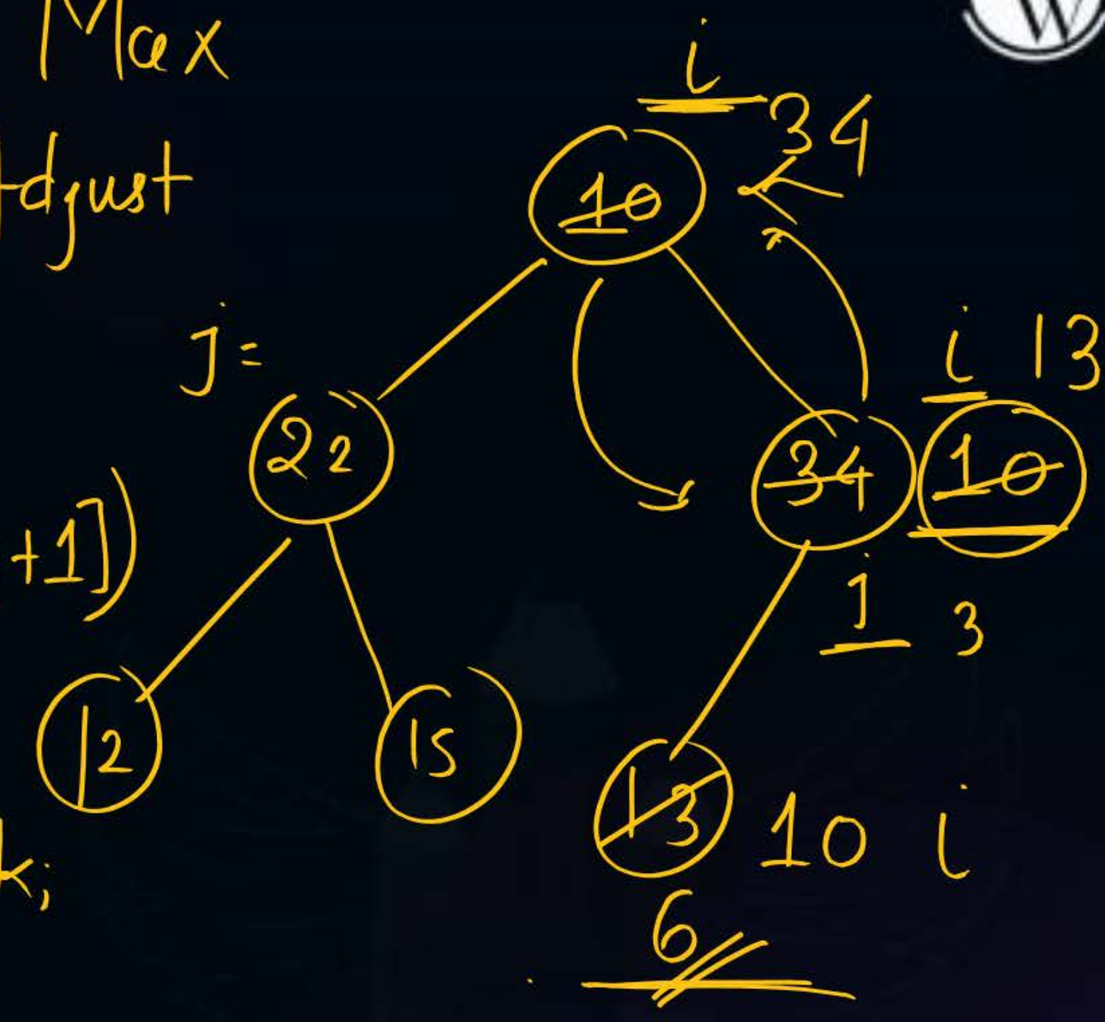
Slide



Topic : Heap

Max Adjust

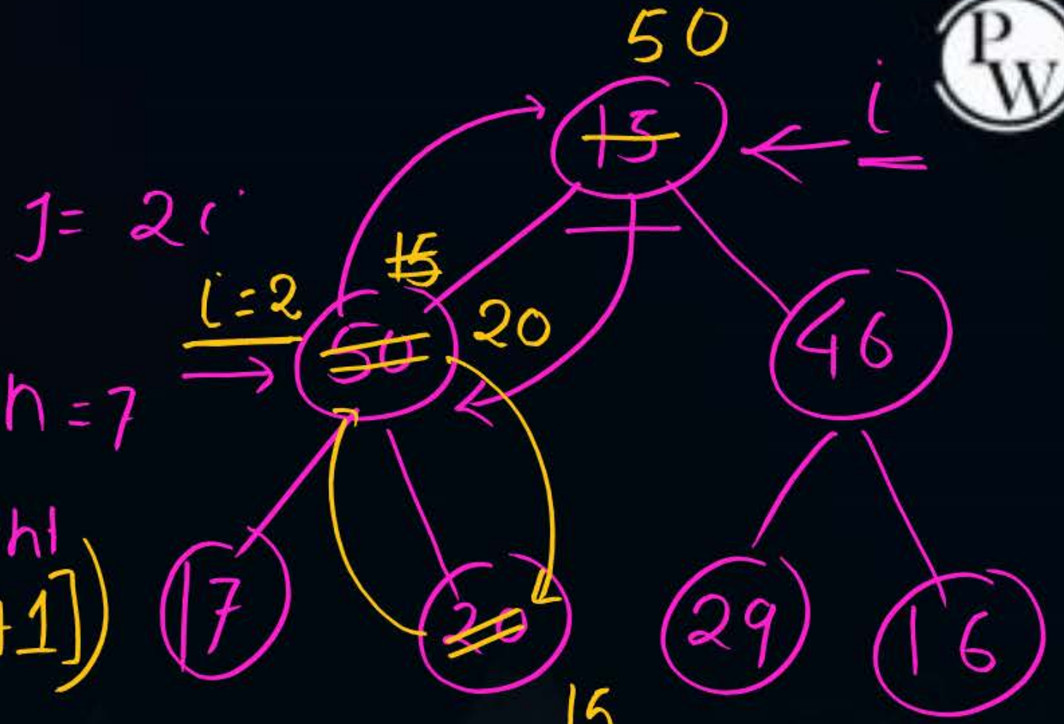
```
Algorithm Adjust(a,n,i) {
    j=2i , x=a[i]
    while(j≤n) {
        if(j<n and a[j]<a[j+1])
            j=j+1;
        if(a[i] > a[j]) break;
        Swap(a[i], a[j])
        i=j, j=2j;
    }
}
```



Slide

Topic : Heap

```
Algorithm Adjust(a, n, i) {
    j = 2i , x = a[i]
    while (j ≤ n) {
        if (j < n and a[j] < a[j+1])
            j = j + 1;
        if (a[i] > a[j]) break;
        Swap(a[i], a[j])
        i = j, j = 2j;
    }
}
```



Slide



Topic : Heap

Counting of Heap

$T(n)$ represent No. of min Heap possible with n nodes

* $n = 1$, 1

$T(1) = 1$

Min Heap



* $n = 2$ 1, 2

$T(2) = 1$



Not CBT

Hence Not Heap

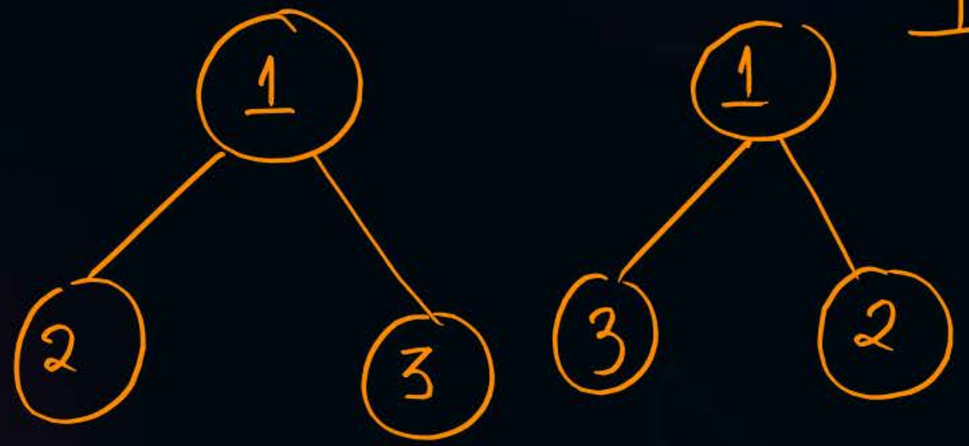
Slide



Topic : Heap

* n=3 1,2,3

T(3) = 2

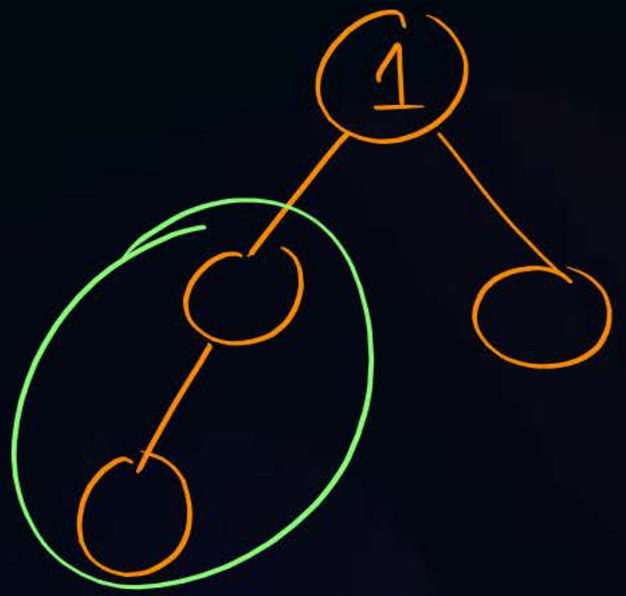


1 CBT structure

Slide

Topic : Heap

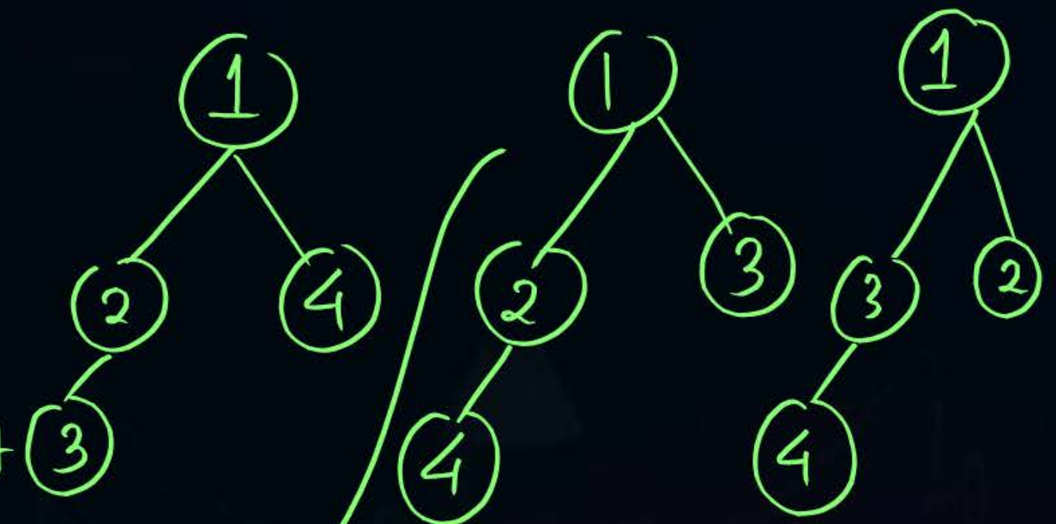
* n=4, 1,2,3,4 T(4) is 3



Node left

3_{C_2} = 3

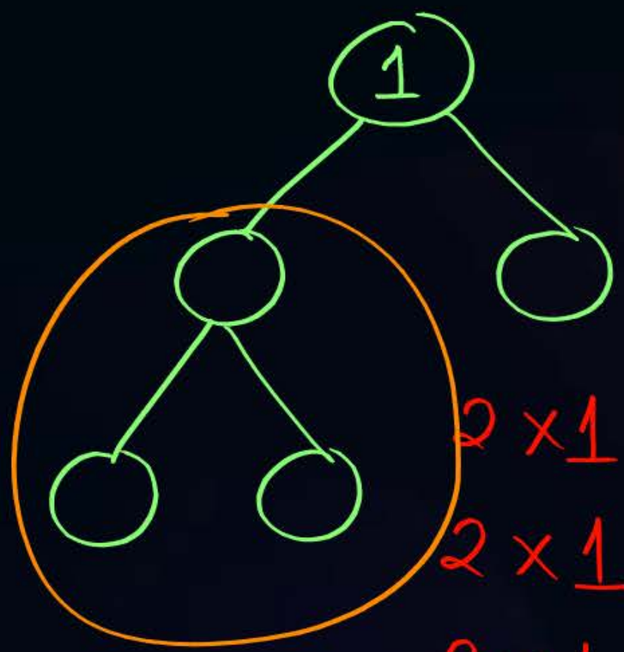
	left	Right
2, 3		4
2, 4		3
3, 4,		2



Slide

Topic : Heap

T(5), 1, 2, 3, 4, 5



$^4C_3$

Left	Right
2, 3, 4	5
3, 4, 5,	2
2, 4, 5	3
2, 3, 5	4

2 × 1 ← 2, 3, 4 | 5

2 × 1 ← 3, 4, 5, | 2

2 × 1 ← 2, 4, 5 | 3

2 × 1 ← 2, 3, 5 | 4

8

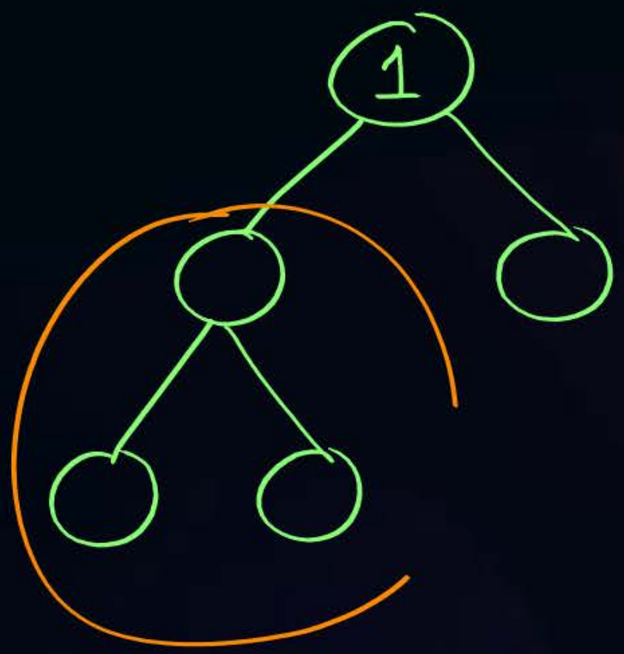


Slide



Topic : Heap

T(5), 1, 2, 3, 4, 5



n=5

k=3

$$\sum_{k=1}^{n-1} {}^{n-1}C_k * T(k) * T(n-1-k)$$

$${}^{4}C_3 * T(3) * T(4-3)$$

$$4 * 2 * 1 = 8$$

Slide



Topic : Heap

formula $T(n) = {}^{n-1}C_K \; T(K) * T(n-1-K)$

1 Draw the CBT structure & find No. of Node in Left Subtree

K

$T(1) = 1 \quad T(2) = 1 \quad T(3) = 2$

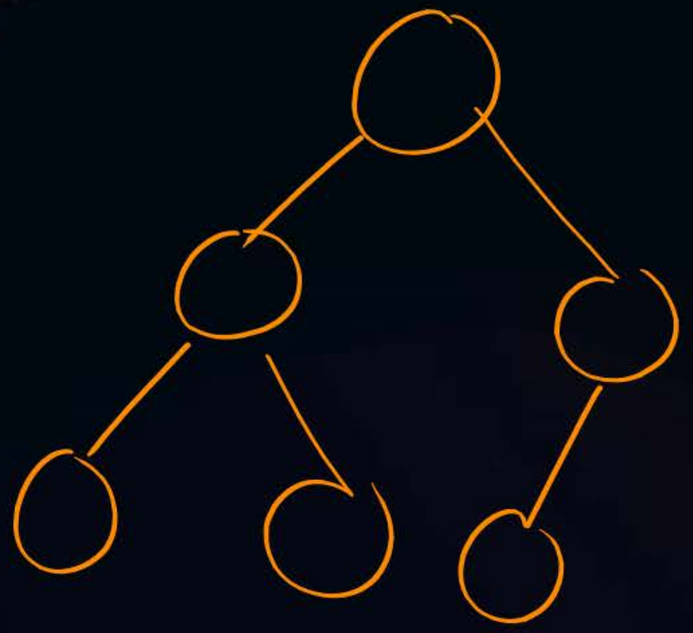
$T(4) = 3$

$T(6)$, 1, 2, 3, 4, 5, 6 ——

Slide



Topic : Heap



$n-1 C_k \; T(k)\; T(n-1-k)$

$5 C_3 \; T(3) * T(2)$

$\frac{5 \times 4^2}{2 \times 1} * 2 * 1$

$10 * 2 = 20$

$n = 6$

$k = 3$

Slide



Topic : Graph

Graph G(V, E)

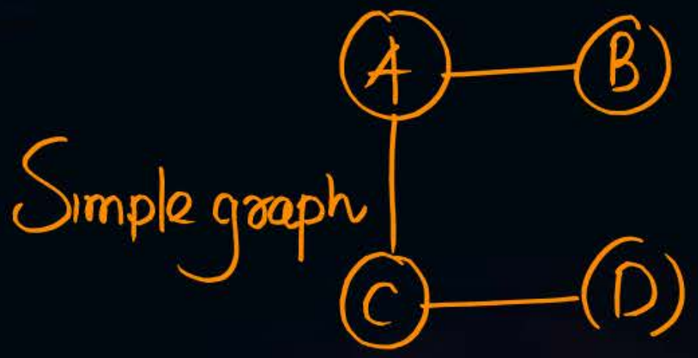
Graph is set of vertices & edges.

Simple graph : No Self loop, No multiple edges between vertices are allowed

Slide



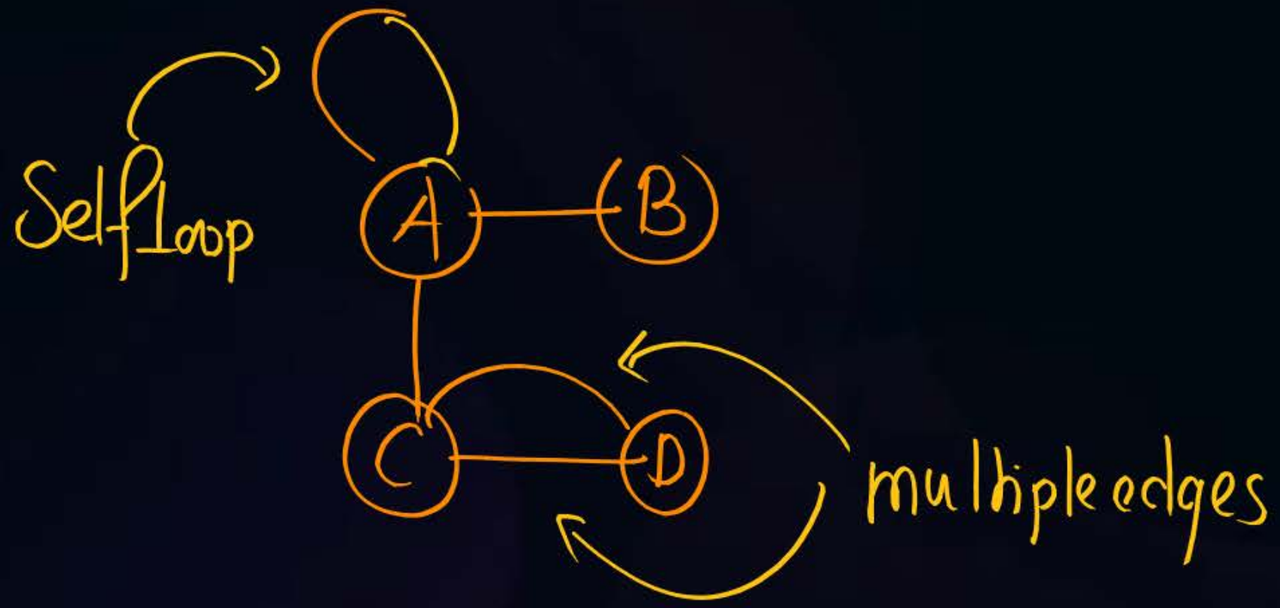
Topic : Graph



Simple graph

$V = \{A, B, C, D\}$

$E = \{(AB), (AC), (CD)\}$



Self loop

multiple edges

Not a simple graph

Slide



Topic : Graph

1 Modeling real world problem

example: * Network connection

* Electric connection — Dijktras
* Road connectivity — (Shortest path)(Spanning tree
* piping connectivity — prims kruskal
* Molicular structure

Slide



Topic : Graph

Representation of graph

1. Adjacency List (Sparse)
2. Adjacency Matrix (Dense)

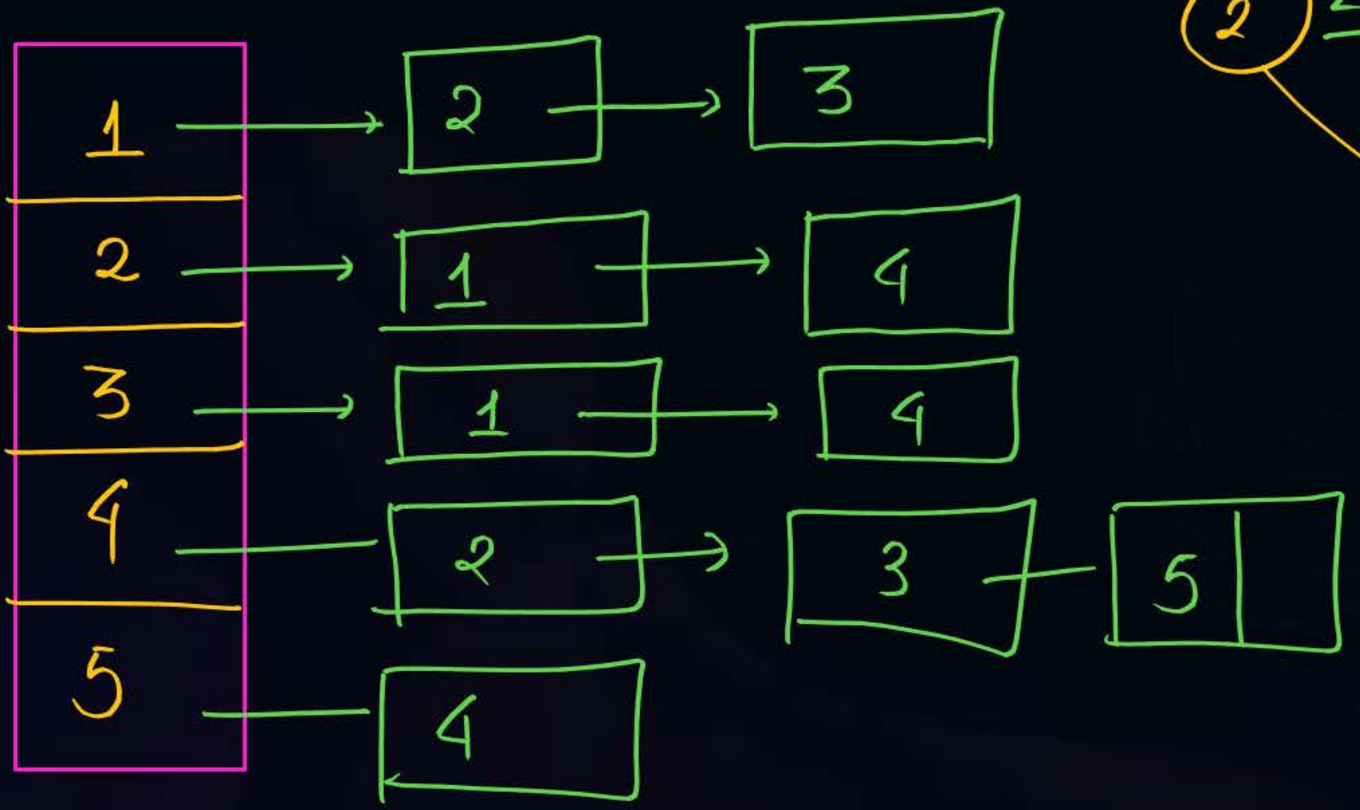


Slide

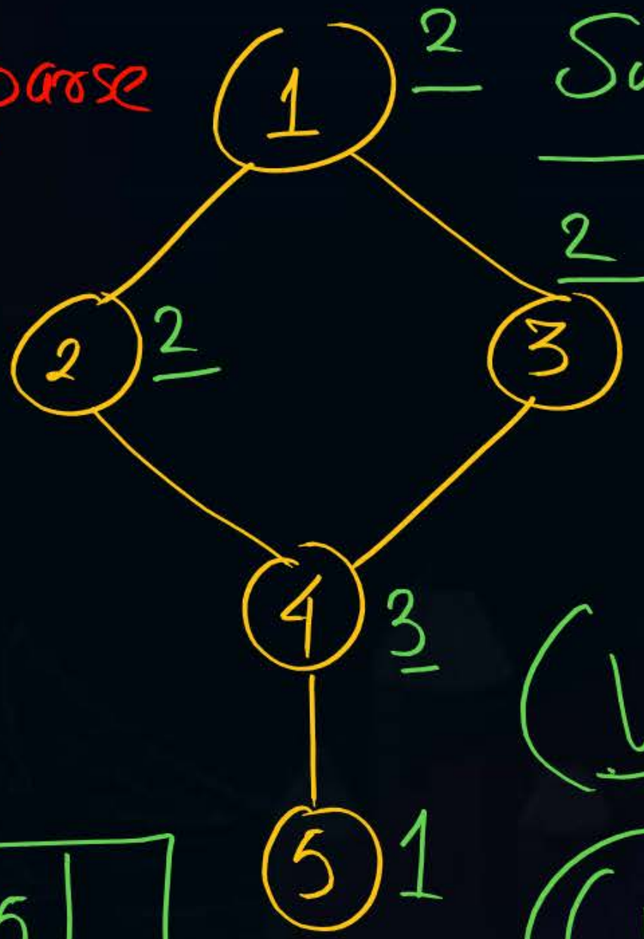


Topic : Graph

Adjacent ≡ Neighbour Node



Sparse



Sum of degree

Space

(V + Sum of degree)

(V+2E)

Slide



Topic : Graph

Adjacency Matrix

M is a Matrix

M[i,j] = 1 ✓

if (i,j) ∈ E

Otherwise 0

	1	2	3	4	5
1	0	1	1	0	0
2	1	0	0	1	0
3	1	0	0	1	0
4	0	1	1	0	1
5	0	0	0	1	0

V^2



Slide



Topic : Graph

Graph Traversal

DFS

Algorithm

BFS

Breadth first

Search

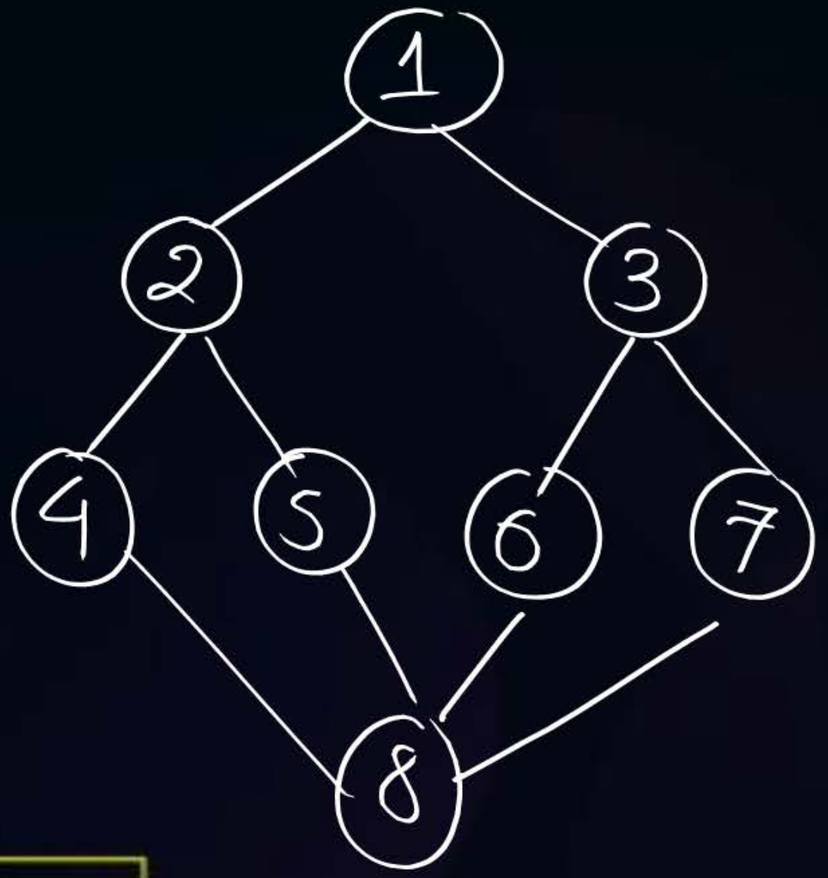


Slide



Topic : Graph

Traversal is process of visiting Nodes of a graph/tree

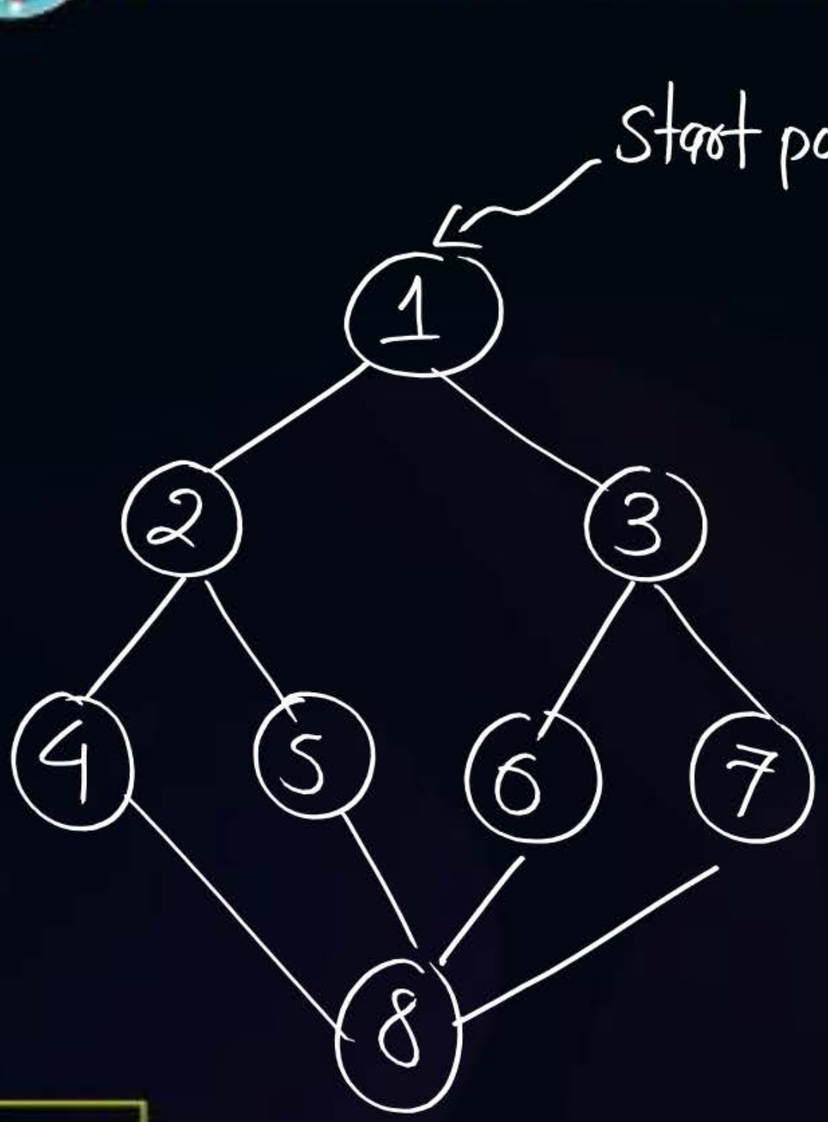


fundamental concept.

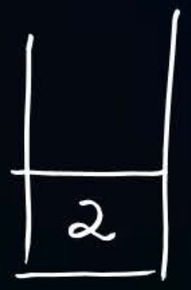
After visiting a Node, Adjacent Node of a vertex is visited. To maintain the order queue data structure is maintained.

Slide

Topic : Graph







Queue

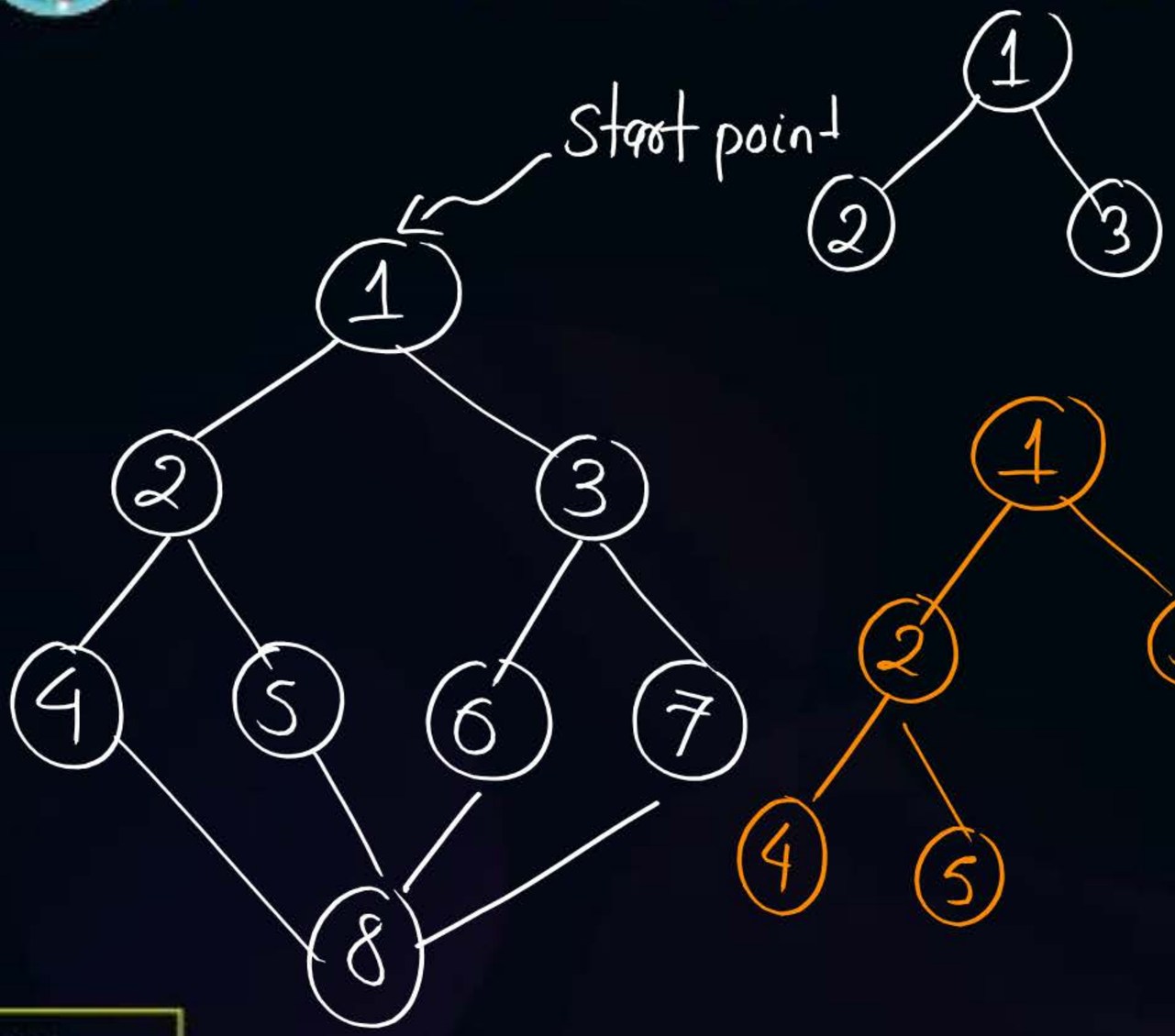


Visit 2 and insert in queue

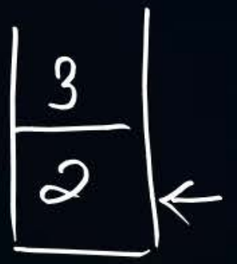
Visit 3 and put them in queue

Slide

Topic : Graph







delete element from queue and visit Adjacent Node (if Not visited)





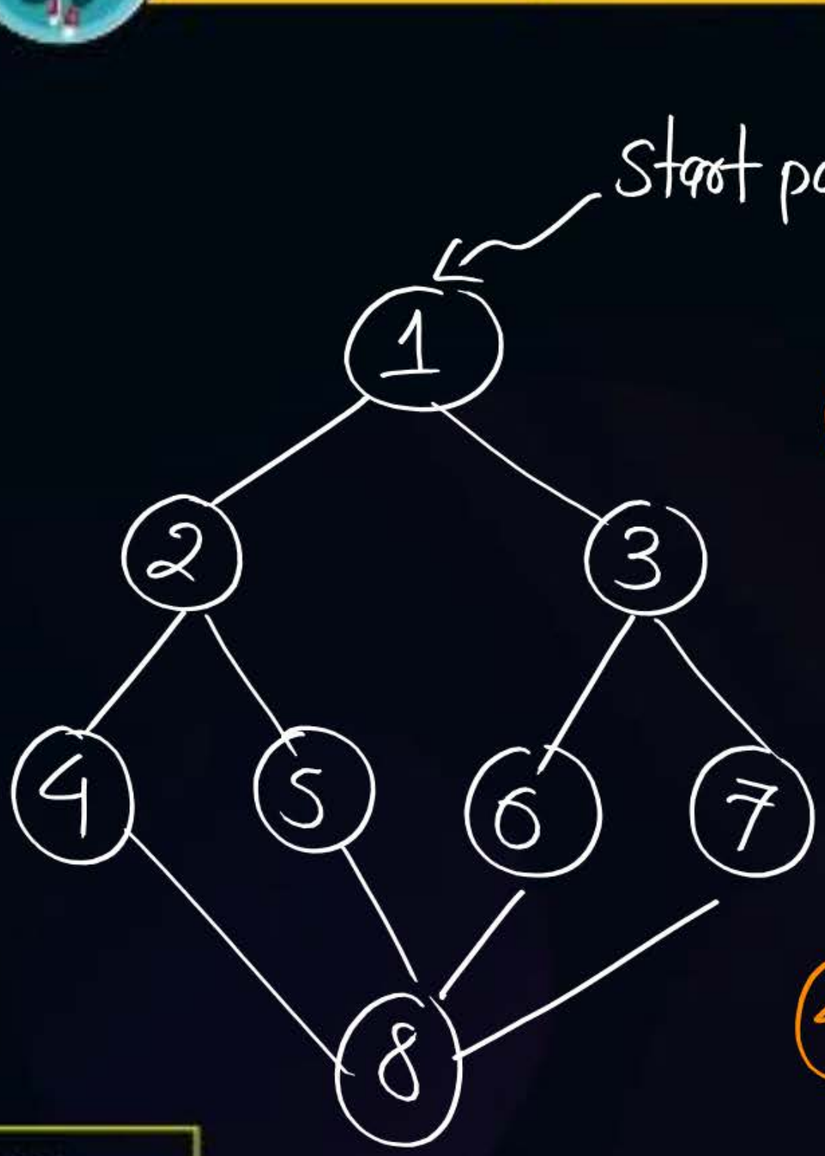
2 deleted

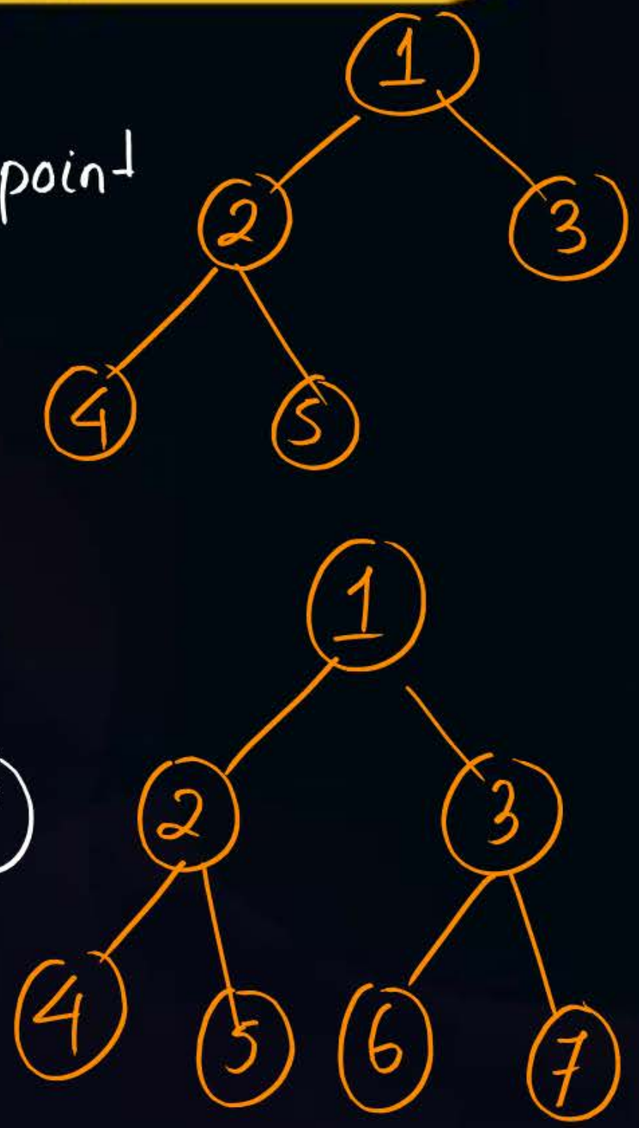
4,5 Inserted

Slide



Topic : Graph



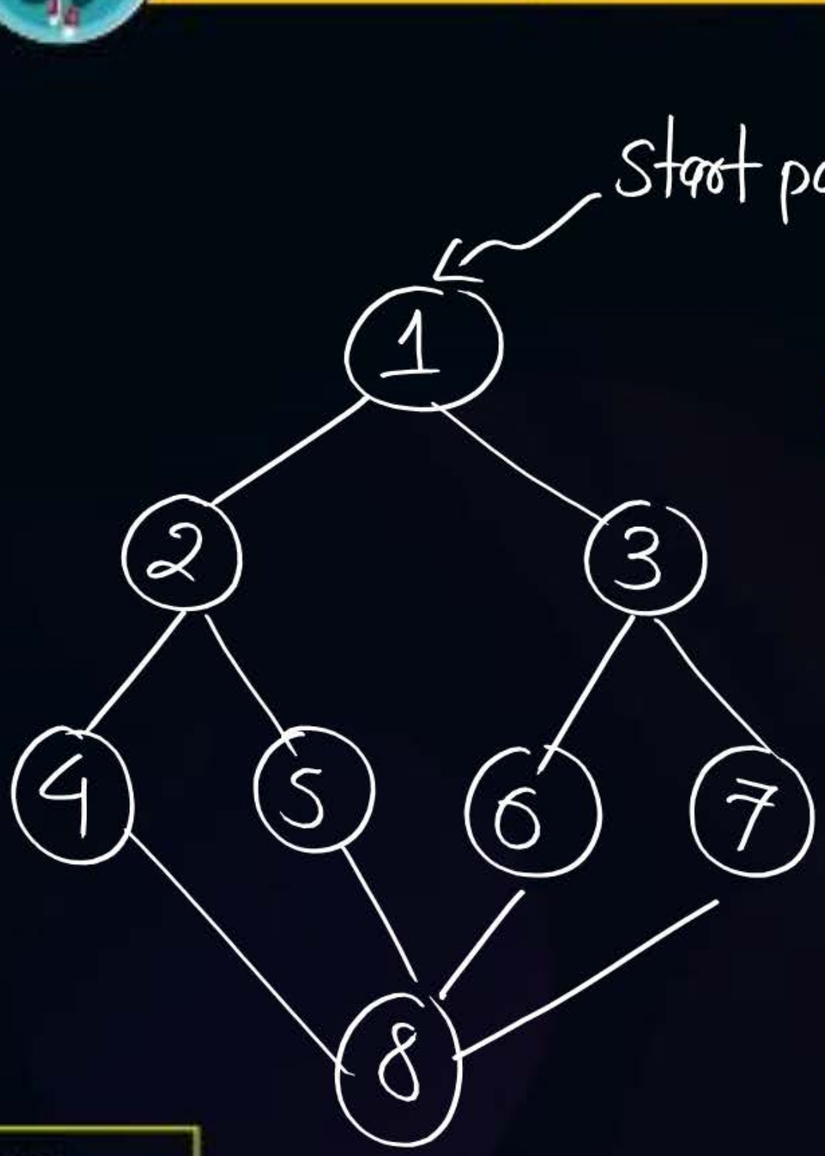




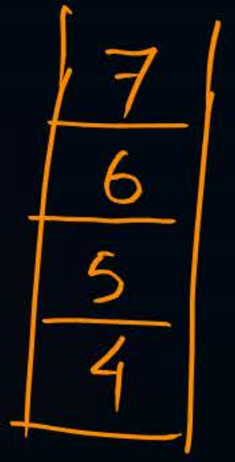
Delete 3
and visit Adjacent
Node 2 Insert
in queue
(if Not visited

Slide

Topic : Graph







Delete 4 and visit Adjacent Node (if Not visited)

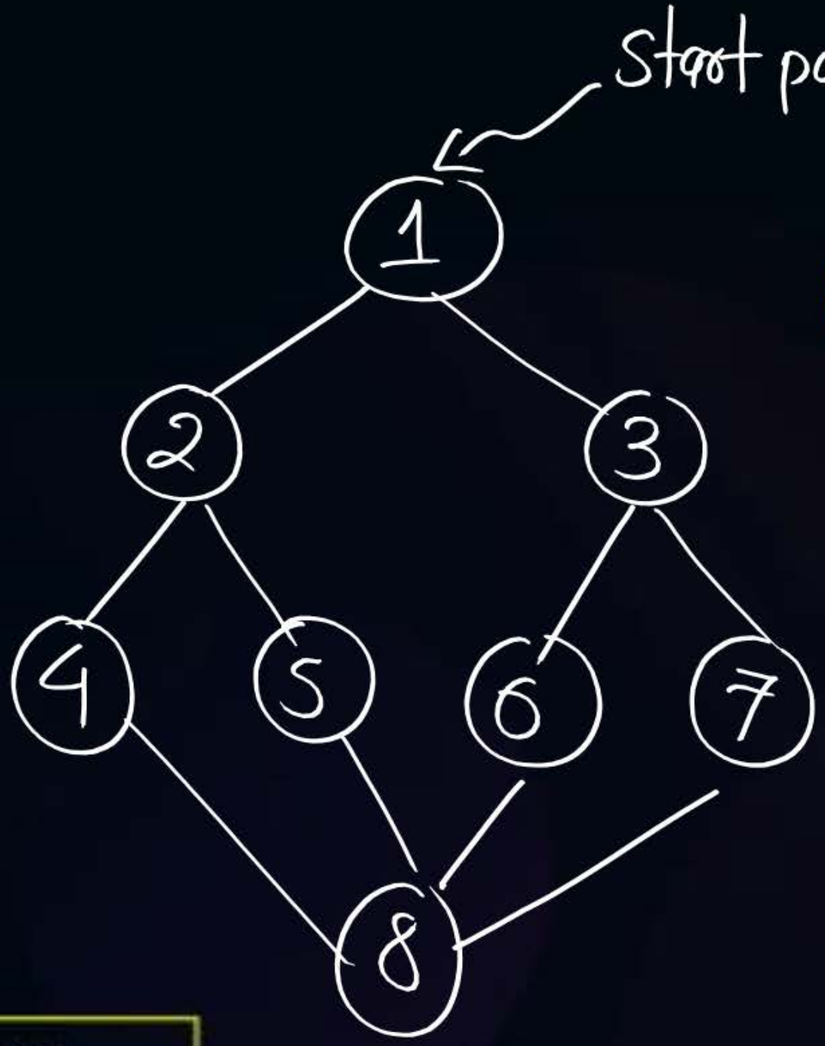


only 8 inserted because 2 is already visited.

Slide



Topic : Graph







because all Nodes are visited

the Node will delete from queue & No New Node will be visited

q empty algorithm stops

Slide



Question

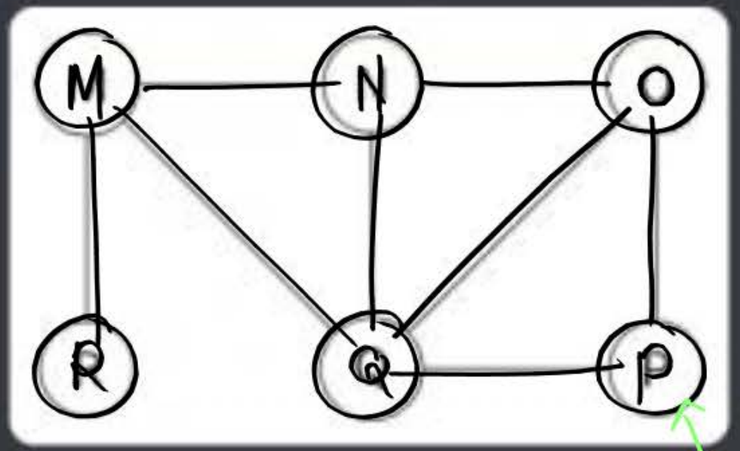
#Q Number of possible with max heap with 12 , 16, 45, 10, 9 is

_____

Slide

Question

The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?



A. MNOPQR
B. NQMPOR
C. QMNROP
D. POQNMR

(A) MNOPQRQ

(B) NQMPORQ

(C) QMNROPQ

(D) POQNMR ✓

Slide



2 mins Summary

Topic — Adjust Algorithm

Topic — Counting of Heap

Topic — Graph Representation

Topic — Graph traversal BFS (repeat)

Topic

Slide

THANK - YOU